

UCLA Undergraduate
RESEARCH WEEK

Empirical Modeling of Solar Influence on Polar Cap Ionospheric Electron Temperature

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Project Overview

What

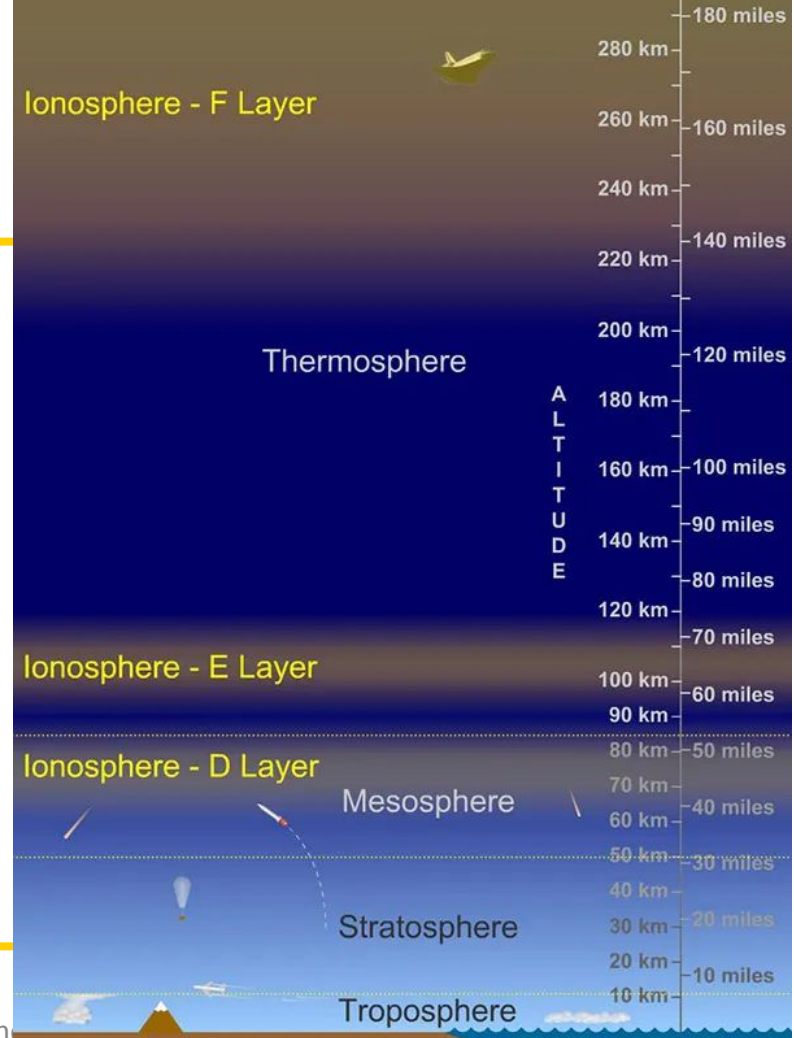
- To develop an empirical model relating electron temperature in the polar cap ionosphere to solar conditions (SZA and F10.7)

Why

- Short term: Examine the influence of SZA and F10.7 on electron temperature
- Long term: Better understand electron heating in the polar cap caused by solar wind

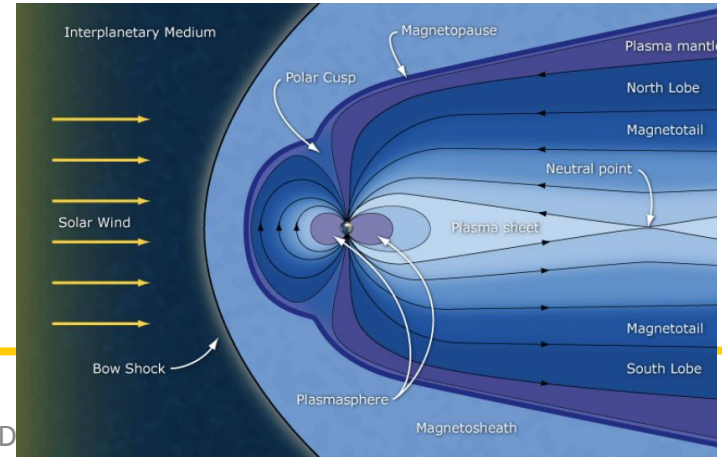
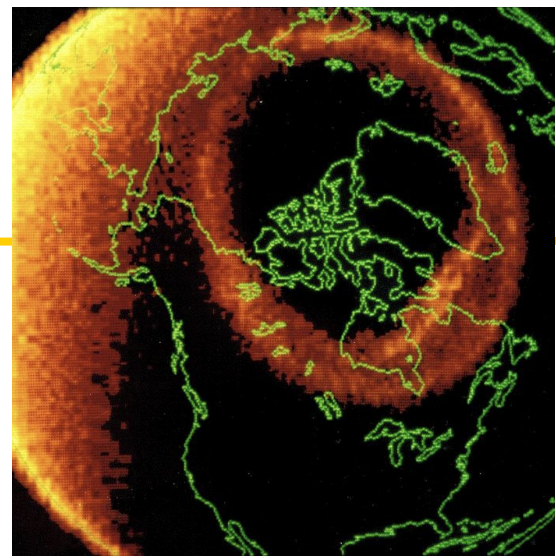
What is the Ionosphere?

- Uppermost region of atmosphere
- Considered to be the edge of Earth and space
- Partially ionized plasma, unlike neutral atmosphere
 - Free electrons
- Investigating the F2-Region (> 220 km)



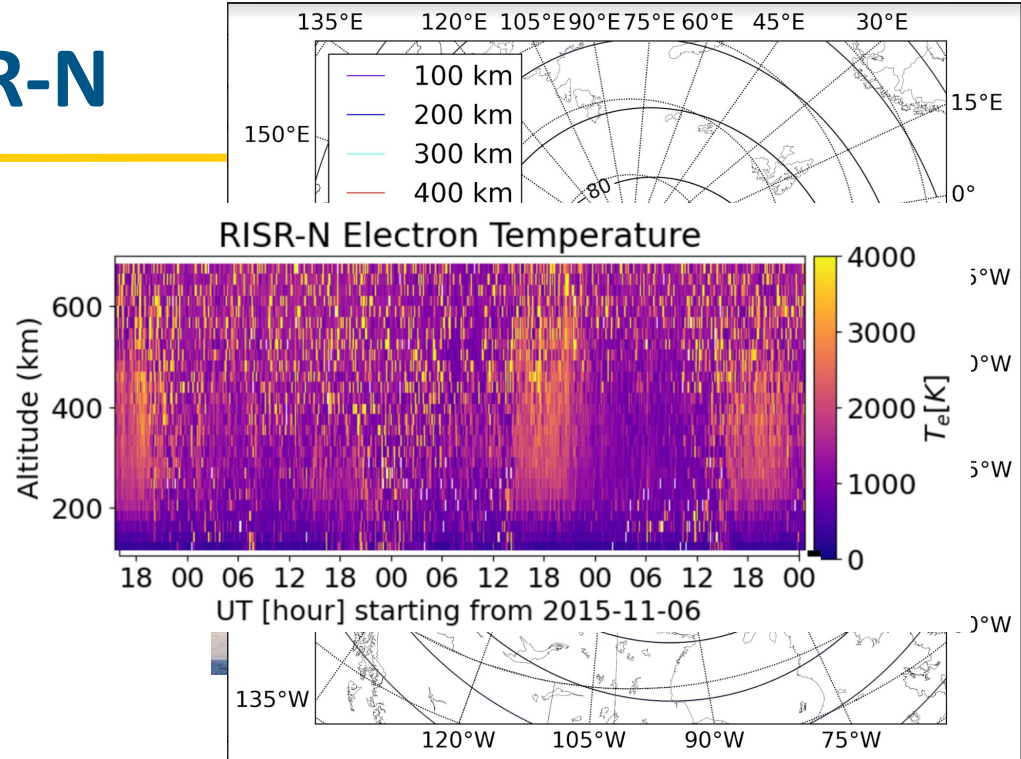
Polar Cap

- Region above 80° magnetic latitude
- Open magnetic field lines connected to solar wind
- Unique heating processes



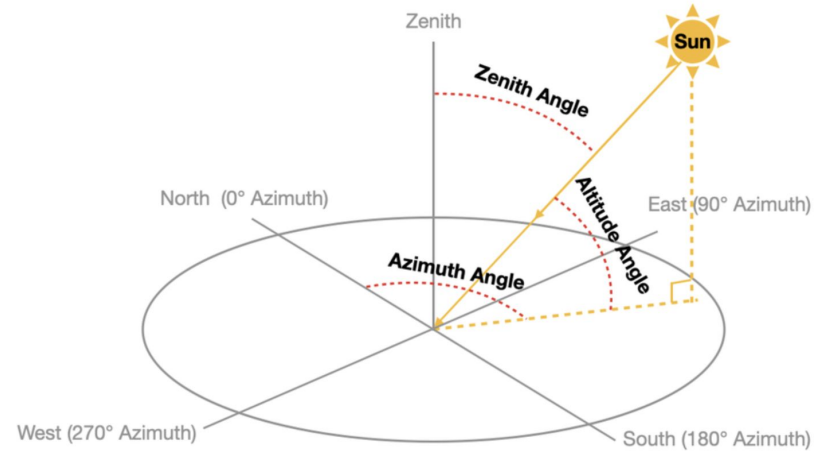
Introduction to RISR-N

- In Resolute Bay, Canada, the RISR-N Facility Collects Radar Data describing Ionospheric Conditions
- Data is processed and can be visualized into 2d color plots, shown on the right



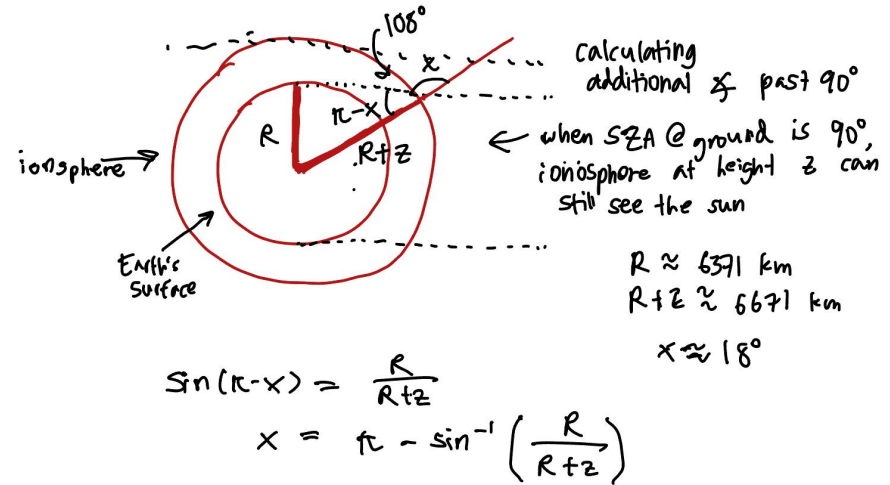
Solar Zenith Angle (SZA)

- Angle between sun and vertical direction
- SZA range (55 - 140 degrees for RISR data)



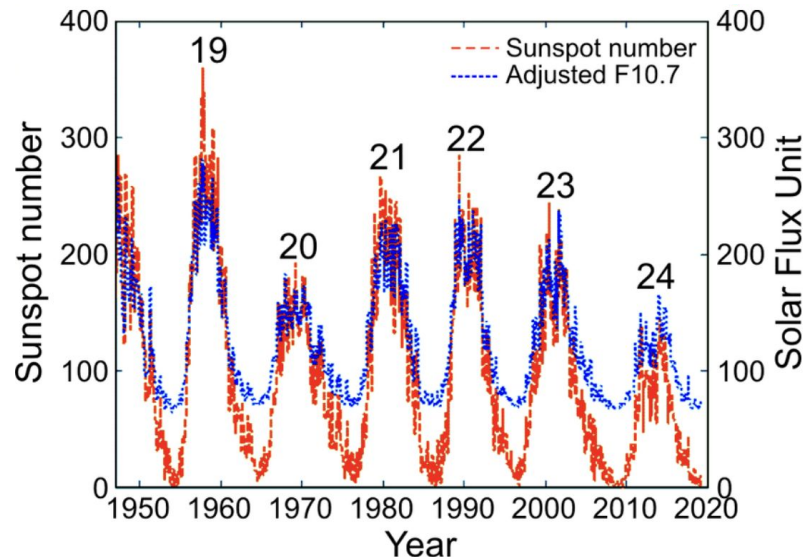
Why 90° SZA is not the same as sunset in the ionosphere

- Due to the Earth's curvature and the high altitude of the F-region, direct sunlight continues to reach the ionosphere
- At 90° SZA on the ground, still heating
- Temperature transition occurring around 95 degrees rather than exactly at 90 degrees



F10.7 Index

- Solar activity proxy using radio emissions
- Varies between 65 - 200 solar flux units
- $10^{-22} \text{ W} * \text{m}^{-2} * \text{Hz}^{-1}$



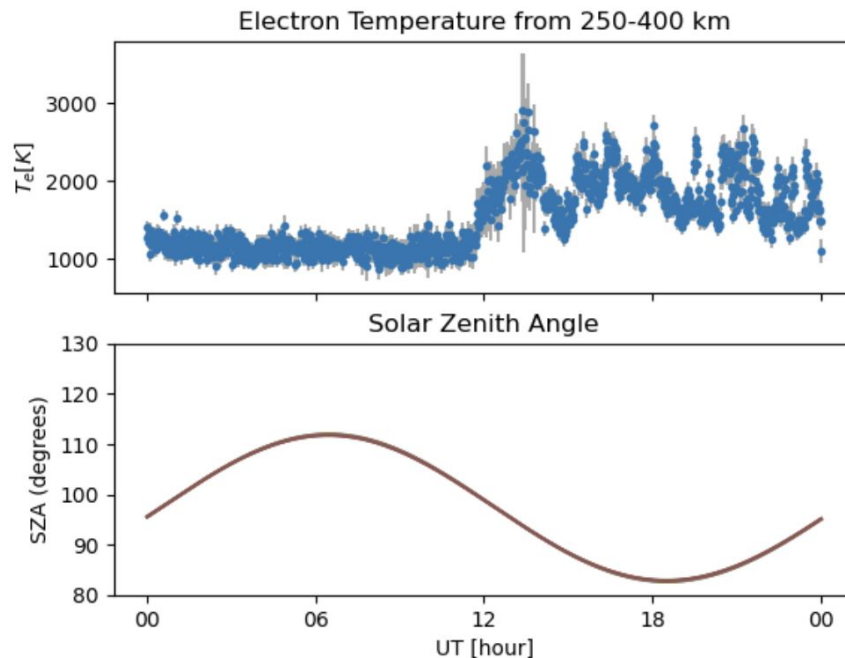
Processing RISR Data

Overview

- 12 years of measurements (2012-2024)
- 250-400 km altitude range

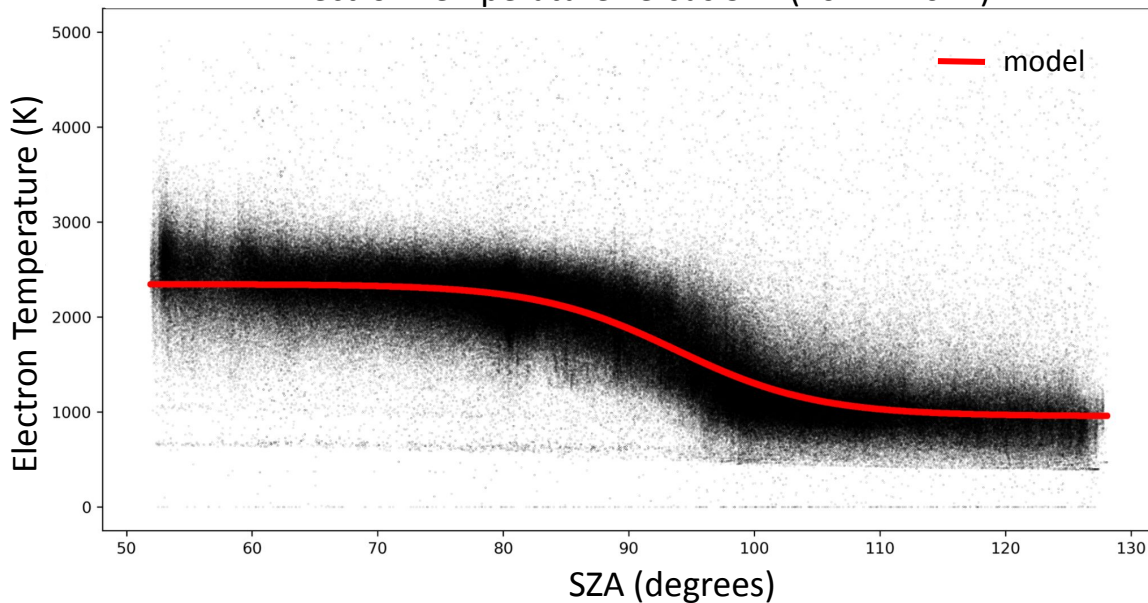
Method

- Removed measurements with high uncertainty
- 0 - 5000 K temperature range
- Calculated median values across altitude range



Electron Temperature vs SZA

Electron Temperature versus SZA (2012 - 2024)



Strong
transition
near 95° SZA

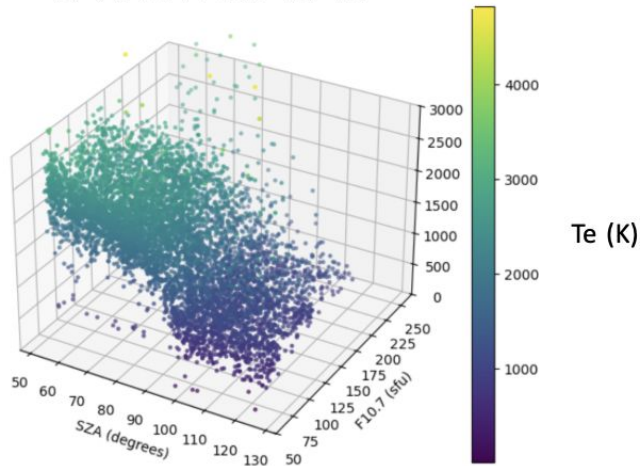
$$Te_{model} = a \tanh\left(\frac{(SZA - b)}{c}\right) + d$$

SZA vs F10.7 vs Te

Te vs. SZA Relationship:

- Strong negative correlation
- Temperature decreases as sun angle increases

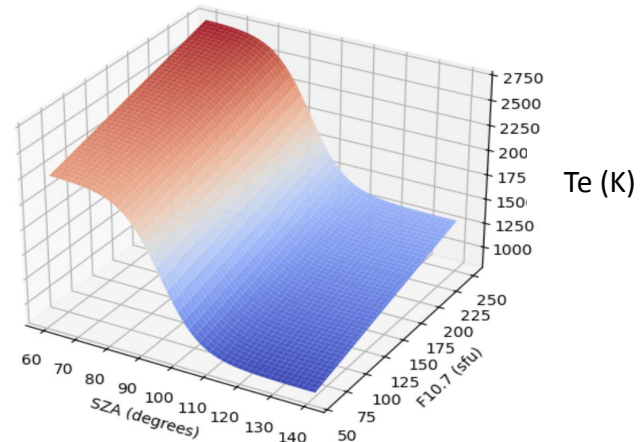
SZA vs F10.7 vs Te



Solar Activity Effect:

- Moderate positive correlation with F10.7
- Additional heating during high solar activity periods

SZA vs F10.7 vs Te Model



Privacy Notice: Do not

$$Te_{model} = a \tanh\left(\frac{(SZA - b)}{c}\right) + d(F10.7) + e$$

Conclusion

Key Findings

- Solar zenith angle is a dominant factor controlling electron temperature

Future Work

- Further isolate relationship between F10.7 and Te
- Investigating the influence of solar wind parameters

Acknowledgements

- Dr. Varney of the AOS Department
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 - This work is supported by the Resolute Bay Observatory which is a major facility funded by the National Science Foundation through cooperative agreement AGS-1840962 to SRI International. RISR-N data is available through the SRI International ISR Database at <https://amisr.com/database/>
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Backup Slides

What does it mean for SZA to be > 90 degrees?

- SZA measures angle between sun and zenith (directly overhead)
- Range: $\sim 50^\circ$ to 140° for RISR
- Sun never directly overhead (0°) at Arctic latitudes
- Minimum SZA ($\sim 50^\circ$) occurs at summer solstice
- Maximum SZA ($\sim 140^\circ$) during winter darkness
- Values $> 90^\circ$ represent sun below horizon

More Info about Model

- Mean squared error (MSE): 110,000 K²
 - Parameters
 - a: -0.69
 - b: 93.8 (solar transition point)
 - c: 11.2
 - d: 0.0026 (F10.7 transition)
 - e: 1.39
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More About the Ionosphere

- Ultraviolet solar radiation ionizes atoms and molecules
 - Contains enough energy to “strip” electrons from a neutral gas
 - Varies with solar activity
 - Results in free electrons and ions (plasma)
 - Electron temperature is the measure of average kinetic energy of free electrons
 - Different from the neutral atmosphere
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